VIDYALANKAR INSTITUTE OF TECHNOLOGY

Department of Computer Engineering · Academic Year 2026–27

Course Insights & Curriculum Mapping (AAP)

Course Code: VEC02 · E-Waste & Environmental Management (EWEM) · 2 Credits

1. Executive Summary: This document establishes the foundational baseline for the VEC02 E-Waste and Environmental Management curriculum. The course is structured as a 2-credit environmental mandate comprising 30 contact lecture hours, mapped comprehensively across 6 core curriculum buckets.

Module #	Curriculum Bucket Title	Hours	CO Alignment
Module 1	E-Waste Genesis & Global Quantification Trends	5 Hrs	CO1 (Understand)
Module 2	Toxicological Hazards, Heavy Metals & Bioaccumulation	5 Hrs	CO2 (Analyze)
Module 3	Recycling Metallurgy: Pyrolysis, Hydrometallurgy & Plasma Arc	6 Hrs	CO3 (Apply)
Module 4	Extended Producer Responsibility (EPR) & E-Waste Rules 2022	5 Hrs	CO4 (Evaluate)
Module 5	Circular Electronics, Eco-Design & Urban Mining Economics	5 Hrs	CO5 (Design)
Module 6	AI in Waste Curation, Logistics & Recommender Pipelines	4 Hrs	CO6 (Synthesize)

2. Evaluation Architecture: Theory-only evaluation consists of In-Semester Assessments (ISA: 40 Marks) focusing on empirical state trackers, elemental blueprints, and policy decks, and End-Semester Assessments (ESE: 60 Marks) culminating in patent concepts, campus activation drives, green building scoring, and IEEE research.